

MASTER'S COMPREHENSIVE EXAMINATION

Part II Applied (90 minutes) March 29th, 2014

Instructions: Attempt any three out of the following four questions on both pages. Mention which three you would like to have graded. Open book and open notes.

1. It is the year 2016 in the United States and Chris Christie (Christie) is running in the Republican primary for the presidential nomination. There are only 3 States left in the voting; Texas, Ohio and Pennsylvania. Christie's statisticians from Rutgers gave him the following probabilities to win:

PROBABILITIES FOR CHRISTIE TO WIN EACH STATE

$P = 0.55$ to Win Ohio, $P = 0.65$ to Win Texas,

$P = 0.61$ to Win Pennsylvania

PROBABILITIES FOR CHRISTIE TO WIN TWO STATES TOGETHER

$P = 0.32$ to {Win Ohio $\cap$ Win Pennsylvania}

$P = 0.32$ to {Win Ohio $\cap$ Win Texas}

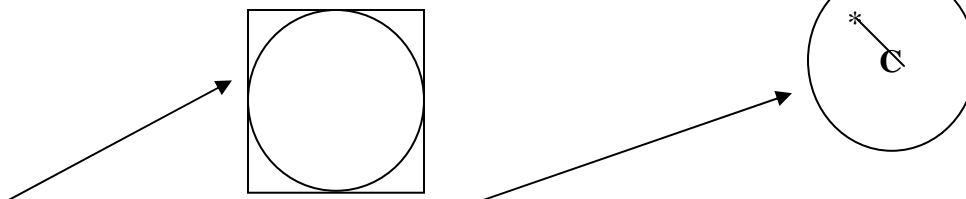
$P = 0.37$ to {Win Pennsylvania $\cap$ Win Texas}

PROBABILITIES FOR CHRISTIE TO WIN ALL THREE STATES

$P = 0.20$ to Win All three States together

- A. Is Christie more likely to win Pennsylvania if he wins Texas or if he loses Texas? No credit if you do not explain your answer correctly.
- B. Texas has 70 delegates, Ohio has 60 and Pennsylvania has 50. A candidate gets all of a State's delegates for winning a State and none otherwise. Christie needs 120 more delegates to win the Republican nomination. What is the expectation and the variance of the probability he will win the Republican nomination?

2. 400 points are independently chosen on a square.



A. If 329 of the 400 points are within the inscribed circle test the null hypothesis of the probability to fall within the circle is random versus the alternative that it is greater than by chance alone relative to area of the circle at $\alpha = 0.05$.

- B. Now consider only the 329 independent points inside the inscribed circle. Suppose the radius of this circle is 1 and the sample mean distance of a point from the center of the circle (C) is $\bar{X} = 0.65$ with a variance of $S^2 = 0.056$ (you can use this sample variance to estimate the variance for the population). Give a large sample test of the two sided null hypothesis on whether or not the points are closer to the center of the circle than they should be by chance alone if the points were randomly distributed inside the circle.

3. A researcher investigates the relation between the circumference of a tree and its age. She picks a random sample of 30 oak tree stumps and 30 pine tree stumps. She codes TreeType = 1 if Pine, TreeType = 0 if Oak. For each tree stump she measures the Circumference (in inches) of the stump, and the Age (in years) of the tree. See computer printout of least-square fit of model below. Use the information in the computer printout to back up your answers.

- Estimate the average circumference of Oak trees that are 95 years old
- Describe the difference between pine and oak trees in the growth rate (inches in circumference per year of age). Is this difference statistically significant?

Response: Circumference

Root Mean Square Error 1.982884

Observations (or Sum Wgts) 60

Parameter Estimates

Term	Estimate	Std Error	t Ratio	Prob> t
Intercept	-1.351916	3.612092	-0.37	0.7096
Age	0.2945472	0.036335	8.11	<.0001
TreeType	-12.57464	4.415263	-2.85	0.0061
Age*TreeType	0.1841789	0.04923	3.74	0.0004

Effect Test

Source	DF	Sum of Squares	F Ratio	Prob>F
Age	1	258.37039	65.7125	<.0001
TreeType	1	31.89132	8.1111	0.0061
Age*TreeType	1	55.03262	13.9967	0.0004

4. A medical researcher seeks to compare different treatments for damaged knees. Each treatment consists of a surgical technique followed by physical therapy. The experiment consists of three factors: Surgical technique (2 levels), Physical Therapy method (3 levels), and Doctors (5 levels). Surgical technique and Physical Therapy method are fixed factors, while Doctor is a random factor. The design is a 2x3x5 factorial design. 150 patients are assigned completely at random, 5 patients to each of the 30 combinations of levels of the factors. The response variable is a measure of recovery; it is normally distributed.

The researcher uses a standard ANOVA program that computes the F-ratios by taking (MS Source of variability / (MS error)). This printout is given below. The researcher sees from the F-ratios in this printout that the Main effects and their interactions are all significant.

a) Compute the correct F-ratios, and state the conclusions.

b) Explain to the researcher your reanalysis, and what if anything can be done.

Standard ANOVA Printout from Computer

Source of Variability	df	MS	F-ratio
Surgical Technique (S)	1	40.00	16.67 *
Physical Therapy approach(P)	2	18.00	7.50 *
Doctor (D)	4	6.00	2.50 *
SP	2	12.00	5.00 *
SD	4	6.00	2.50 *
PD	8	5.80	2.42 *
SPD	8	4.20	1.75 NS
Error	<u>120</u>	<u>2.40</u>	
Total	149		